

## IA Mathematics for Natural Sciences B:

### Examples Sheet 2 - Triple products, vector area, and polar coordinates

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Questions marked with a (\*) are difficult and should not be attempted at the expense of the other questions.

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#### Vector products: perpendicular projections

1. Draw a diagram showing that the magnitude of the component of  $\mathbf{v}$  perpendicular to  $\mathbf{u} \neq \mathbf{0}$  is  $|\mathbf{v} \times \hat{\mathbf{u}}|$ . Show further that the projection of  $\mathbf{v}$  perpendicular to  $\mathbf{u}$  is  $\hat{\mathbf{u}} \times (\mathbf{v} \times \hat{\mathbf{u}})$ . Using these results:
  - (a) Show that an equation of a line passing through the point  $\mathbf{a}$ , in the direction  $\mathbf{v} \neq \mathbf{0}$ , is  $\mathbf{r} \times \mathbf{v} = \mathbf{m}$ , where  $\mathbf{m}/|\mathbf{v}|$  is the closest point on the line to the origin. Find the equations of the lines  $L_1, L_2$  from Sheet 1 Question 6 in the form  $\mathbf{r} \times \mathbf{v} = \mathbf{m}$ ; hence, find their shortest distances to the origin.
  - (b) Explain why an equation of the plane through  $\mathbf{a}$  with normal  $\mathbf{n} \neq \mathbf{0}$  is  $\hat{\mathbf{n}} \times ((\mathbf{r} - \mathbf{a}) \times \hat{\mathbf{n}}) = \mathbf{r} - \mathbf{a}$ .
2. By considering appropriate perpendicular projections, show that:
  - (a) the shortest distance from a point  $\mathbf{p}$  to a line passing through a point  $\mathbf{a}$  with direction vector  $\mathbf{v}$  is given by  $|(\mathbf{p} - \mathbf{a}) \times \hat{\mathbf{v}}|$ , and the closest point on the line to the point is  $\mathbf{p} - \hat{\mathbf{v}} \times ((\mathbf{p} - \mathbf{a}) \times \hat{\mathbf{v}})$ .
  - (b) the shortest distance from a point  $\mathbf{p}$  to a plane passing through a point  $\mathbf{a}$  with normal  $\mathbf{n}$  is given by  $\sqrt{|\mathbf{p} - \mathbf{a}|^2 - |(\mathbf{p} - \mathbf{a}) \times \hat{\mathbf{n}}|^2}$ .

Hence, use the vector product to find the shortest distance from the point  $P = (1, 0, 1)$  to the planes  $P_1, P_2$  and the lines  $L_1, L_2$  from Sheet 1 Questions 6 and 7.

#### Vector and scalar triple products

3. State *Lagrange's formula* for the *vector triple product*  $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ ; now, learn it off by heart. Give an example of three vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  such that  $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) \neq (\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ . Using Lagrange's formula:
  - (a) prove *Jacobi's identity*,  $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0}$ ;
  - (b) verify that the equation of the plane in Question 1(b) is equivalent to  $(\mathbf{r} - \mathbf{a}) \cdot \mathbf{n} = 0$ .
4. Consider the equation  $\mathbf{a} \times \mathbf{r} + \lambda \mathbf{r} = \mathbf{c}$ , for the variable  $\mathbf{r}$ , where  $\lambda \neq 0$  and  $\mathbf{a}, \mathbf{c} \in \mathbb{R}^3$  are constant.
  - (a) By taking the scalar product of the equation with  $\mathbf{a}$ , show that  $\lambda(\mathbf{r} \cdot \mathbf{a}) = \mathbf{c} \cdot \mathbf{a}$ .
  - (b) By taking the vector product of the equation with  $\mathbf{a}$ , and using (a), find the equation's solution.
  - (c) Does the equation have any solutions when  $\lambda = 0$  and  $\mathbf{a} \cdot \mathbf{c} \neq 0$ ? What if  $\lambda = 0$ , and  $\mathbf{a} \cdot \mathbf{c} = 0$ ? Find the solutions, if they exist, in each case. [Hint: what does (a) tell you about  $\mathbf{r} \cdot \mathbf{a}$  in these cases?]
5.
  - (a) Solve  $2\mathbf{r} + \hat{\mathbf{n}} \times \mathbf{r} + \hat{\mathbf{n}}(\hat{\mathbf{n}} \cdot \mathbf{r})^2 = \mathbf{a}$  for  $\mathbf{r}$ , where  $\hat{\mathbf{n}}$  is a unit vector, and  $\hat{\mathbf{n}} \cdot \mathbf{a} = -1$ ;
  - (b) Solve  $\mathbf{r} = \mathbf{a} - (\mathbf{b} \cdot \mathbf{r})\mathbf{b} - \mathbf{b} \times \mathbf{r}$  for  $\mathbf{r}$ , where  $\mathbf{a}, \mathbf{b}$  are constant vectors.
6.
  - (a) Define the *scalar triple product*  $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$  of three vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$ . With a diagram, show that the scalar triple product  $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$  represents the *signed* volume of the parallelepiped formed by  $\mathbf{a}, \mathbf{b}, \mathbf{c}$ .
  - (b) Hence, explain why: (i) the scalar triple product is antisymmetric on odd permutations, and symmetric on even permutations; (ii) three vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  form a basis if and only if  $[\mathbf{a}, \mathbf{b}, \mathbf{c}] \neq 0$ .
  - (c) Let  $\mathbf{a} = (0, \frac{1}{2}, \frac{1}{2})$ ,  $\mathbf{b} = (\frac{1}{2}, 0, \frac{1}{2})$  and  $\mathbf{c} = (\frac{1}{2}, \frac{1}{2}, 0)$ . Compute  $[\mathbf{a}, \mathbf{b}, \mathbf{c}]$ ,  $[\mathbf{b}, \mathbf{a}, \mathbf{c}]$  and  $[\mathbf{b}, \mathbf{c}, \mathbf{a}]$ , and comment on how they are related in light of (b)(i). Do the vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  form a basis?
7. Simplify the scalar triple products  $(\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} + \mathbf{c}) \times (\mathbf{c} + \mathbf{a})$  and  $(\mathbf{a} \times \mathbf{b}) \cdot [(\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a})]$ .

8. (a) Given the vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  are orthonormal, derive expressions for the coefficients  $\alpha, \beta, \gamma$  in the formula  $\mathbf{d} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$ . Hence express  $(2, 3, 4)$  in terms of the basis  $\{(1, 1, 0), (1, -1, 0), (0, 0, 1)\}$ .
- (b) If instead the vectors  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  are *not* orthonormal, derive expressions for the coefficients  $\alpha, \beta, \gamma$  in the formula  $\mathbf{d} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$ . Hence express  $(1, 1, 1)$  in terms of the basis  $\{(1, 2, 1), (0, 0, 1), (2, -1, 1)\}$ .
- (c) Assuming  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  form a basis, we define the *reciprocal basis* to  $\mathbf{a}, \mathbf{b}, \mathbf{c}$  to be the set of vectors:

$$\mathbf{A} = (\mathbf{b} \times \mathbf{c}) / [\mathbf{a}, \mathbf{b}, \mathbf{c}], \quad \mathbf{B} = (\mathbf{c} \times \mathbf{a}) / [\mathbf{a}, \mathbf{b}, \mathbf{c}], \quad \mathbf{C} = (\mathbf{a} \times \mathbf{b}) / [\mathbf{a}, \mathbf{b}, \mathbf{c}].$$

Show that  $\mathbf{A} \cdot \mathbf{a} = \mathbf{B} \cdot \mathbf{b} = \mathbf{C} \cdot \mathbf{c} = 1$ , and  $\mathbf{A} \cdot \mathbf{b} = \mathbf{A} \cdot \mathbf{c} = \mathbf{B} \cdot \mathbf{a} = \mathbf{B} \cdot \mathbf{c} = \mathbf{C} \cdot \mathbf{a} = \mathbf{C} \cdot \mathbf{b} = 0$ . Find the reciprocal basis to the basis in part (b), and explain how to use it to find the coefficients  $\alpha, \beta, \gamma$ .

### Vector area

9. (a) Define the *vector area*  $\mathbf{A}$  of a connected, flat surface that has area  $A$  and unit normal  $\hat{\mathbf{n}}$ . What is the convention that is normally used when choosing the direction of the normal?
- (b) Determine the vector areas of the: (i) parallelogram spanned by  $\mathbf{a}, \mathbf{b}$ ; (ii) triangle spanned by  $\mathbf{a}, \mathbf{b}$ .
- (c) The velocity  $\mathbf{u} = (1, 0, 1)^T$  of a fluid is uniform everywhere. Compute the volume passing the triangle spanned by  $\mathbf{a} = (0, 0, 1)^T, \mathbf{b} = (0, 1, 0)^T$  per unit time. [*This is called the velocity flux.*]
10. (a) Define the vector area  $\mathbf{A}$  of a connected, non-flat surface composed piecewise of flat surfaces of areas  $A_1, \dots, A_k$  with unit normals  $\hat{\mathbf{n}}_1, \dots, \hat{\mathbf{n}}_k$  respectively. What do we require about the normals?
- (b) Find the vector area  $\mathbf{S}$  of the surface  $S$  made up of the two triangles  $OAB, BCO$ , where  $A = (1, 0, 0), B = (1, 1, 1), C = (0, 2, 0)$ . Is  $|\mathbf{S}|$  equal to the surface area of  $S$ ?
11. Give a very general outline of how the concept of vector area might be extended to *curved* surfaces  $S$ ; hence, explain why we expect the vector area of a *closed* surface to be  $\mathbf{0}$ . Thus, find the vector areas of:
- (a) a square-based pyramid with vertices  $(0, 0, 0), (2, 0, 0), (2, 2, 0), (0, 2, 0), (1, 1, 1)$ , with base removed;
- (b) the unit hemisphere  $x^2 + y^2 + z^2 = 1, z \geq 0$ .
12. (a) Let  $\mathbf{S}$  be the vector area of the surface  $S$ . Prove that the area of the projection (or 'shadow') of the surface  $S$  onto the plane with unit normal  $\hat{\mathbf{m}}$  is  $|\mathbf{S} \cdot \hat{\mathbf{m}}|$ .
- (b) Using the result from part (a), compute the area *and* the vector area of the projection of the square with vertices  $(0, 0, 0), (2, 0, 0), (2, 2, 0)$  and  $(0, 2, 0)$  onto a plane with normal  $\mathbf{m} = (0, -1, 1)$ .

### Polar coordinate systems

13. Draw convincing diagrams defining plane polar, cylindrical, and spherical coordinates. Hence, find the cylindrical polar and spherical polar coordinates of the point  $(3, 4, 5)$ .
14. (a) In 2D Cartesians, a circle is specified by  $(x - 1)^2 + y^2 = 1$ . Find its equation in polars.
- (b) In 3D Cartesians, a sphere is specified by  $(x - 1)^2 + y^2 + z^2 = 1$ . Find its equation in sphericals.
15. Let  $a > 0$  be a constant. Describe the following loci:
- (a) (i)  $\phi = a$ ; (ii)  $r = \phi$ , in plane polar coordinates;
- (b) (i)  $z = a$ ; (ii)  $r = a$ ; (iii)  $r = a$  and  $z = \phi$ , in cylindrical polar coordinates;
- (c) (i)  $\theta = a$ ; (ii)  $\phi = a$ ; (iii)  $r = a$ ; (iv)  $r = \theta = a$ , in spherical polar coordinates.
16. Consider a point with position vector  $\hat{\mathbf{n}}$  on the unit sphere  $S$  centred on the origin.
- (a) Show that  $\hat{\mathbf{n}} = (\sin(\theta) \cos(\phi), \sin(\theta) \sin(\phi), \cos(\theta))^T$ , where  $(r, \theta, \phi)$  are polar coordinates.
- (b) Show that the vector area  $d\mathbf{S}$  of a small patch near  $\hat{\mathbf{n}}$ , subtending a small angle  $d\theta$  in the  $\theta$  direction, and a small angle  $d\phi$  in the  $\phi$  direction, is given *approximately* by  $d\mathbf{S} = \hat{\mathbf{n}} \sin(\theta) d\theta d\phi$ .
- (c) (\*) Hence, use integration to verify that the vector area of the unit sphere is  $\mathbf{0}$ .